Can 10-K Vocabulary Distinguish Intel and NVIDIA

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Industry and Research Question

I chose Intel and NVIDIA because both companies have impressed me for different reasons. I remember Intel's advertisements from around 2003, while NVIDIA recently caught my attention with its performance in the AI chip sector. Currently, both companies remain active in the semiconductor industry (NAICS 3344) and compete in the data center chip sector, despite differences in their business operations. Then, after comparing the two companies, I became curious whether the differences in their businesses are also reflected in the language used in their annual reports.

Therefore, this study aims to explore whether it is possible to distinguish between texts written by Intel and NVIDIA by analyzing the vocabulary used in Items 7, 7A, and 8 of their 10-K filings.
I used filings submitted from 2020 to 2023 for training and those submitted in 2024 for testing.

Also, I compared the lexical differences with financial metrics to assess the practical utility of these texts for business or investment decisions. Identifying the company behind a given text does not directly determine which stock is a better investment choice; that determination requires careful consideration of factors such as growth, profitability, valuation, and risk.

Market Context

Table 1

Yahoo Finance Market Snapshot for Intel and NVIDIA

Metric	Intel (INTC)	NVIDIA (NVDA)
Market capitalization (USD billions)	635.55	5,514.69
Trailing price to earnings	Unavailable	28.87
Beta	2.23	2.22
Revenue growth	25.40%	105.90%
Return on equity	-10.715%	117.21%
Debt to equity (Yahoo raw field)	49.00	16.97
Five-year cumulative return	142.36%	1,005.90%

Note: Data from Yahoo Finance was retrieved at approximately 02:07 UTC on October 1, 2026 (10:07 p.m. Eastern Daylight Time on September 30, 2026). The five-year cumulative return was calculated using adjusted closing prices from October 1, 2021, to September 30, 2026. The debt-to-equity ratio is shown as reported by Yahoo Finance.

Table 1 presents a September 30, 2026, market snapshot, whereas the text classifier uses filings submitted in 2020 to 2024. NVIDIA reports higher revenue growth (105.90% versus 25.40%) and return on equity (117.21% versus −10.715%). Intel's trailing P/E was unavailable in the Yahoo response, so this table does not support a like-for-like P/E comparison.

Q1 Corpus Construction and Section Comparison

The SEC submissions index identified ten annual reports, one per company and filing year. Their Items 7, 7A, and 8 produced 30 nonempty section rows and 2,719 clean, overlapping sentence chunks. A balanced, evenly spaced subset of 1,624 chunks populated the 5,000-term TF-IDF matrix and both classifiers; the complete chunk collection was also saved to Drive. Intel's 10-K cross-reference required locating its MD&A, market-risk discussion, and financial statements by their report headings; NVIDIA's Item 8 cross-reference led to the statements in Part IV, Item 15. The extraction log shows all 30 sections, including the shorter Item 7A rows, and no missing section. Figures 1 and 2 show the filing balance and chunk lengths.

Figure 1

SEC 10-K filing counts by company and filing year

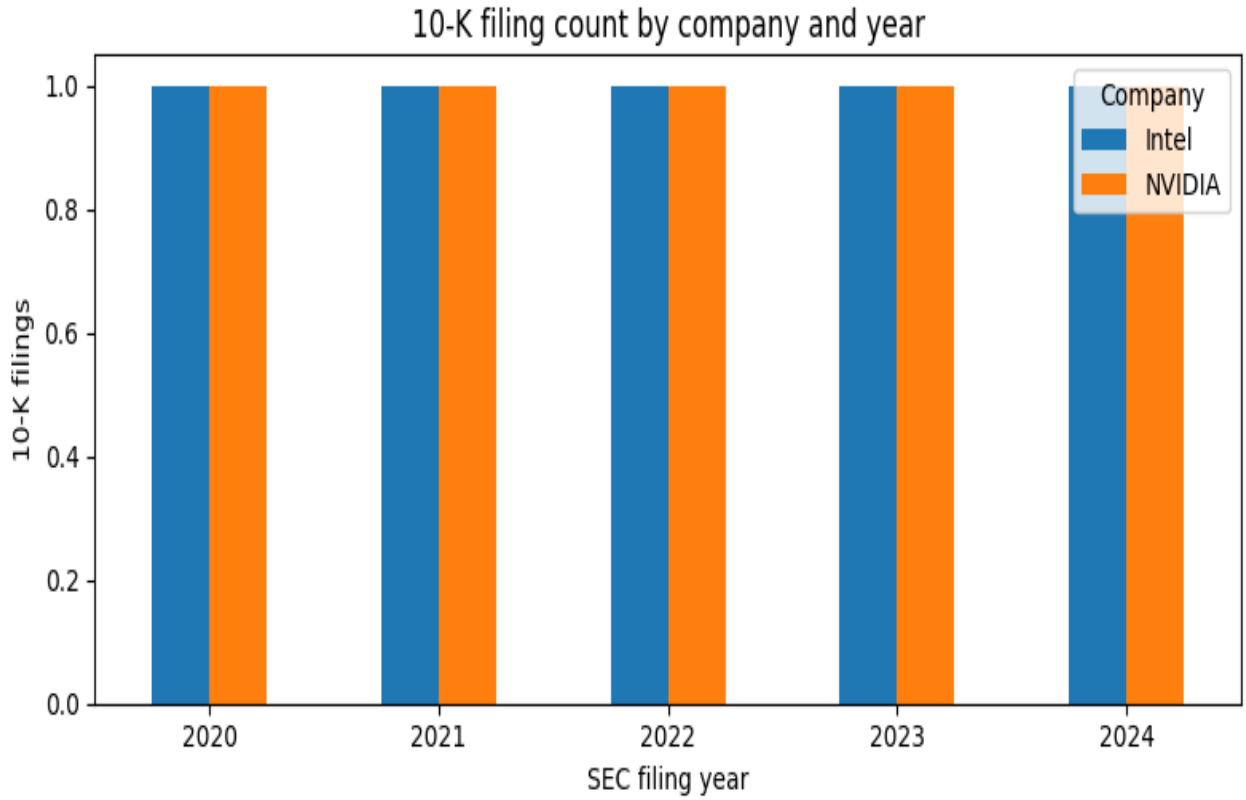
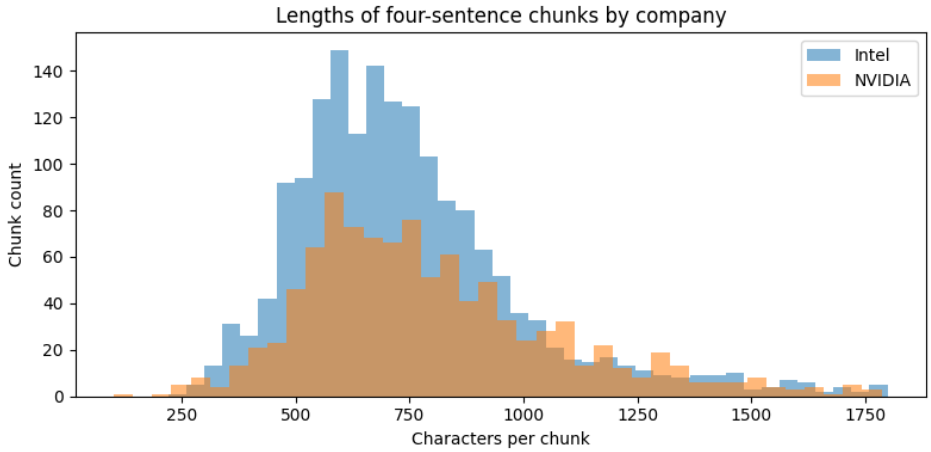


Figure 2

Character lengths of clean chunks by company



For this section comparison, I trained a separate TF-IDF vectorizer and PyTorch classifier for each Item, using filings submitted in 2020–2023 for training and those submitted in 2024 for testing.

Table 2

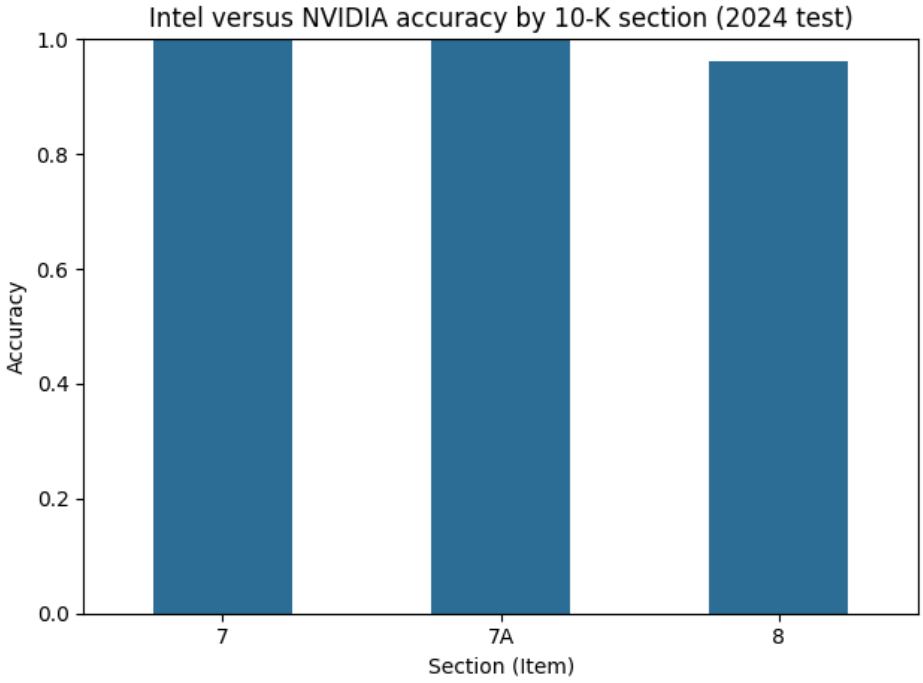
Classification Accuracy by 10-K Section on the 2024 Test Year

Section	Training chunks	Test chunks	Accuracy
7	573	151	1.0000
7A	83	17	1.0000
8	640	160	0.9625

Here items 7 and 7A tied for the highest held-out accuracy at 1.0000, compared with 0.9625 for Item 8 (Table 2 and Figure 3). Item 7 was tested on 151 chunks, whereas Item 7A had only 17; the perfect Item 7A score is therefore less informative than its percentage alone suggests. For these filings, management discussion and market-risk language distinguished the firms more consistently than the standardized financial-statement material. Classification accuracy does not measure future stock returns.

Figure 3

Held-out classification accuracy by 10-K section



Q2 Classifier and Similarity Baseline

Table 3

Accuracy of Two Text Methods on the Same 2024 Test Set

Method	Accuracy
Cosine similarity nearest-chunk baseline	0.984756
PyTorch weighted BoW classifier	0.978659

The cosine nearest-chunk baseline correctly classified 323 of 328 test chunks (0.984756), compared with 321 of 328 (0.978659) for the linear PyTorch model (Table 3). The classifier lagged by 0.006098 accuracy units, or about 0.61 percentage points. Both methods use the same TF-IDF vocabulary and a 2024 filing-year holdout. This small difference gives no evidence that the trained linear model adds useful contextual understanding over vocabulary similarity. Figure 4 shows common high-weight terms, while Figure 5 shows the terms with the largest between-

company advantage. In Figure 6, Item 8 sections are more similar to each other than Items 7 or 7A across all five years.

Figure 4

Top twenty mean TF IDF terms for each company

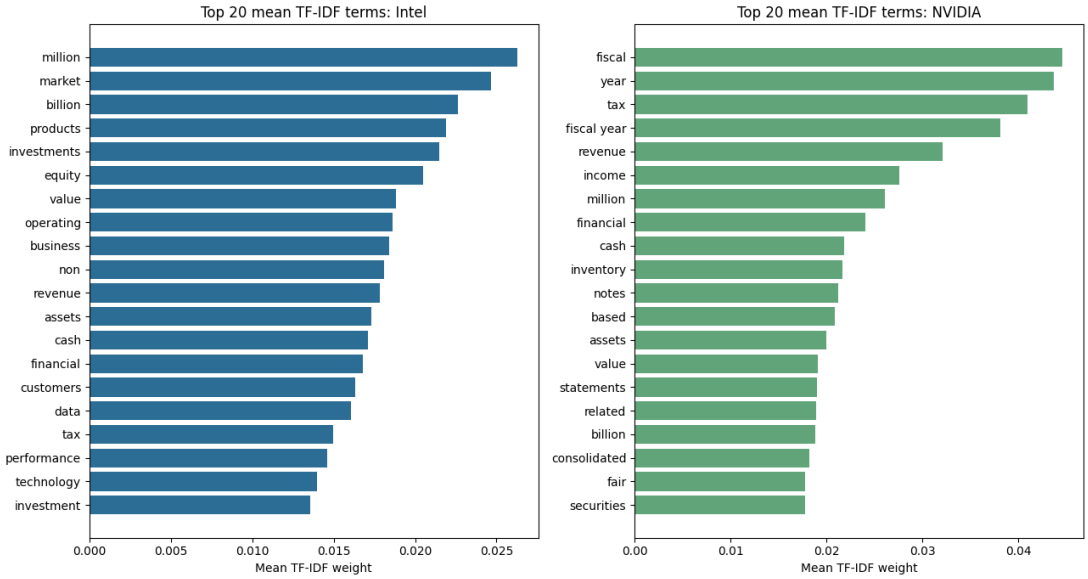


Figure 5

Largest between company mean TF IDF advantages

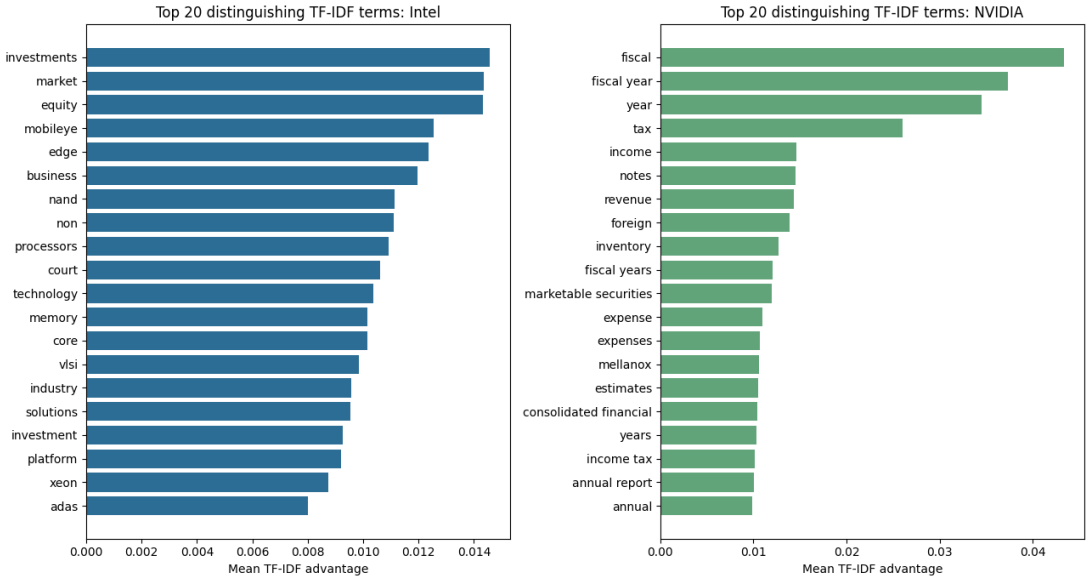


Figure 6

Cosine similarity of matched filing sections by year

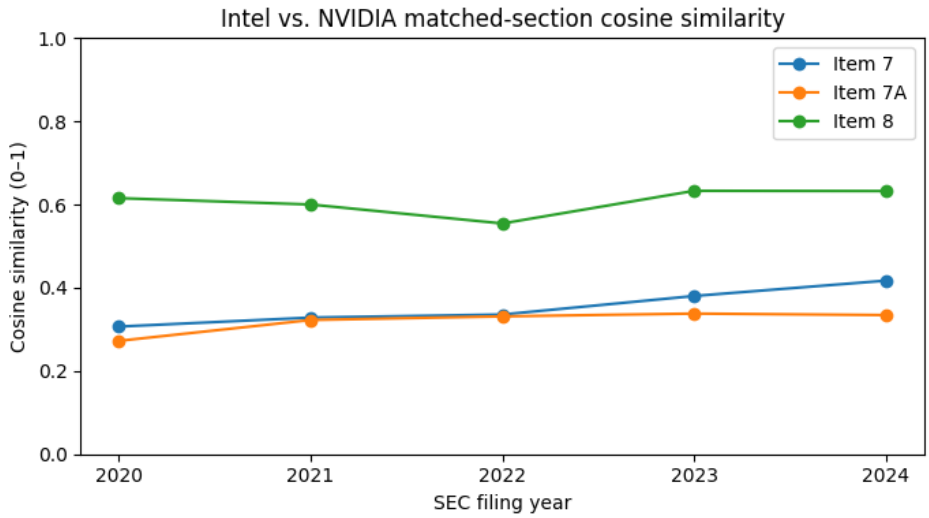
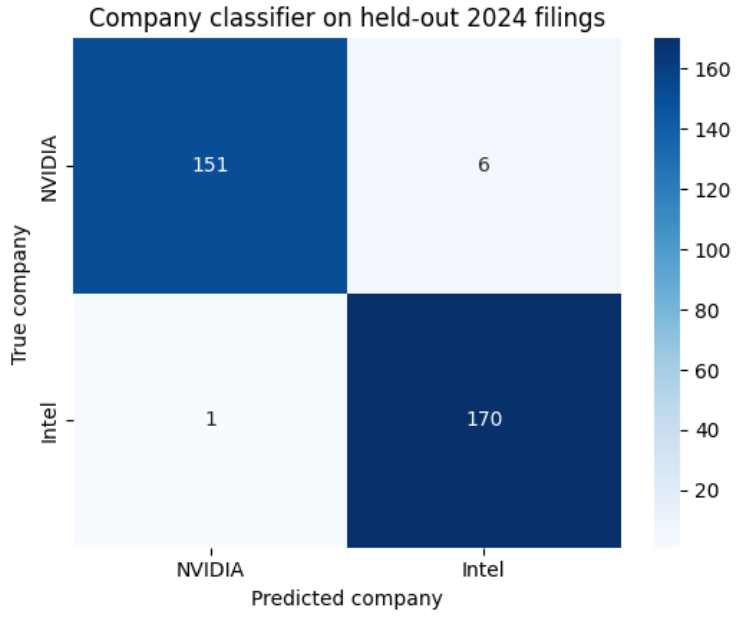


Figure 7

Confusion matrix for the 2024 test chunks



The errors discussed below come from the combined model trained on all three Items, rather than the separate models used in Table 2.

Reliability and Failure Cases

Seven of 328 held-out chunks were misclassified: six NVIDIA excerpts were labeled Intel and one Intel excerpt was labeled NVIDIA (Figure 7). Table 4 shows five of these actual errors; the following interpretations describe plausible failure modes rather than proven causes.

Table 4

Wrong Company Predictions for Manual Review

Chunk ID	True	Predicted	Excerpt
INTC_2024_8_324	Intel	NVIDIA	In 2009, we recorded and paid an EC-imposed fine that was subsequent…
NVDA_2024_7_42	NVIDIA	Intel	To the extent that a customer requires products covered by the licen…
NVDA_2024_7_120	NVIDIA	Intel	Return rights for certain stocking distributors for specific product…
NVDA_2024_7_132	NVIDIA	Intel	We allocate the total transaction price to each distinct performance…
NVDA_2024_8_180	NVIDIA	Intel	Fair value is based upon observable inputs in an inactive market and…

Note. Excerpts are shortened for the page; full text is in the saved error CSV.

NVDA_2024_8_180 discusses investments and equity securities but is predicted as Intel; both terms favor Intel in Figure 5. Shared accounting vocabulary can therefore mislead a word-count model. Second, NVDA_2024_7_42 describes U.S. export-license restrictions affecting AI sales, yet is also labeled Intel. The model may miss the geopolitical context of a newer risk. Third, INTC_2024_8_324 describes an EC fine, VLSI litigation, and a terminated Tower deal but is labeled NVIDIA. Unusual legal events may differ from learned product language. Only 17 Item 7A chunks and one filing year were tested, so the observed accuracies do not establish wider reliability.

Q3 Words, Numbers, and Business Recommendation

Keywords alone would not lead me to the same portfolio decision as the financial numbers. “Revenue” favors NVIDIA in Figure 5, and Table 1 reports 105.90% revenue growth versus Intel’s 25.40%. NVIDIA’s “income” term directs attention to its 117.21% return on equity versus Intel’s −10.715%. These are links to investigate, not claims that word frequency caused the ratios. Intel-side “processors” and “mobileye” describe products absent from the Yahoo table, while NVIDIA-side “fiscal year” may partly reflect filing style. I would provisionally weight NVIDIA more heavily in a two-stock comparison because of growth and profitability but limit the tilt: both betas exceed 2 and the market snapshot follows the 2020–2024 filing corpus. I would next check current 10-Ks, valuation, and export-control exposure before making an investment decision.

References

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